MDP Project — What We Did

Air-quality machine learning across 204 Indian monitoring stations (1.18 million station-hours, 2025).

1. Gaps we covered

- **Honest testing (no data leakage):** we tested by time and by unseen stations, so our accuracy (R^2 ~ 0.25–0.30) is real, not the inflated 0.41.
- **Cause vs. correlation:** we did not treat 'things move together' as proof of cause; only one link (NO → ozone) held up.
- **Weather confounding:** we re-checked every causal link with weather added; the weak ones disappeared.
- **Grouping is not source-finding:** clustering pollutants is NOT source apportionment (that needs chemical data we don't have).
- **Model numbers vs. real policy:** we proved the 'optimizer' only moved numbers in a model, so we don't over-claim real policy.
- **Units & reproducibility:** we verified units officially (found a ppb vs µg/m³ mix in CO, NO_2, SO_2) and released all data + code.

2. Languages & tools we used

Python — main programming language

XGBoost — predicts pollution levels

Optuna / NSGA-II — finds trade-off policies

Matplotlib — draws the charts

Google Cloud (Dataproc) — runs on the cloud

PySpark — handles very large data

DirectLiNGAM — finds cause-effect links

SHAP — explains the model

Git / GitHub — saves & shares code

LaTeX — types the research paper

3. Reference papers we used

1. Rautela & Goyal (2024), *Scientific Reports* — deep learning PM2.5, India
2. Malhotra et al. (2024), *Journal of Big Data* — AI air-pollution review
3. Qi et al. (2024), *Environ. Research Letters* — causal PM2.5↔ozone
4. Wang et al. (2025), *Environ. Sci. & Ecotechnology* — causal + XAI ozone
5. Utku et al. (2025), *Atmosphere* — CNN-RNN PM2.5 forecasting
6. Ye et al. (2026), *Atmos. Chemistry & Physics* — SHAP ozone drivers
7. Lakra et al. (2024), *Atmospheric Environment* — PMF PM2.5, Lucknow/Kanpur
8. Just et al. (2020), *Atmos. Environment* — spatial-CV data leakage
9. Grange & Carslaw (2019), *Sci. Total Environment* — weather normalisation
10. Kaufman et al. (2012), *ACM TKDD* — data leakage, definition
11. Paatero & Tapper (1994), *Environmetrics* — PMF source apportionment
12. Shimizu et al. (2011), *JMLR* — DirectLiNGAM causal discovery
13. Chen & Guestrin (2016), *KDD* — XGBoost prediction model
14. Lundberg & Lee (2017), *NeurIPS* — SHAP explainability
15. Deb et al. (2002), *IEEE Trans. Evol. Comp.* — NSGA-II optimizer
16. Pearl (2009), Cambridge Univ. Press — causality foundations
17. Sillman (1999), *Atmospheric Environment* — ozone–NOx–VOC chemistry
18. Seinfeld & Pandis (2016), Wiley — atmospheric chemistry textbook